Full configuration interaction in a hydrogenic basis on the S^3 label set: Multi-electron atoms via relational lattice indexing

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(Dated: March 12, 2026)

We implement full configuration interaction (FCI) for multi-electron atoms in a hydrogenic basis on the S^3 label set (n, l, m): the physical Hamiltonian is the diagonal hydrogenic one-body term plus two-electron integrals assembled by Slater–Condon rules from exact R^k radial integrals and Gaunt angular coupling coefficients. For helium, the ground-state energy with the physical Hamiltonian converges monotonically to $-2.8476\,\mathrm{Ha}$ (1.93% error versus exact) at $n_{\max} = 5$, above the Hartree–Fock limit: the fixed-exponent hydrogenic bound-state basis omits the continuum. For lithium, the energy converges to $-7.4007\,\mathrm{Ha}$ (1.03% error) at $n_{\max} = 5$. Against strong baselines the comparison is unfavourable: the helium energy stays above the Hartree–Fock limit, which correlation-consistent Gaussian bases pass already at cc-pVDZ; only the one-function STO-3G basis is bettered. For beryllium (4 electrons), the solver achieves $E = -14.563\,\mathrm{Ha}$ (0.71% error) at $n_{\max} = 4$ with 487,635 determinants—the first four-electron calculation in this framework. For lithium hydride (4 electrons, first heteronuclear molecule), the R-independence of each atom's one-electron block makes the potential-energy surface monotonically attractive with no equilibrium geometry, and it does not return to the separated-atom energy at large R, so no binding energy is extracted. An excitation-driven direct CI algorithm following Knowles and Handy [1] replaces the $O(N_{\mathrm{SD}}^2)$ pairwise determinant loop with $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$ assembly, achieving a 45× speedup for lithium. The Hamiltonian is assembled via a relational lattice index that maps Slater determinants to sparse matrix entries without evaluating continuous four-center integrals, achieving >99% sparsity at the largest basis sizes tested. These results establish hydrogenic-basis FCI on the S^3 label set as a working, if basis-limited, route for few-electron systems.

I. INTRODUCTION

Full configuration interaction provides the exact solution to the electronic Schrödinger equation within a given one-particle basis, but its application to atoms and molecules is limited by two computational bottlenecks: the factorial growth of the determinant space and the $O(N^4)$ cost of evaluating two-electron repulsion integrals over Gaussian basis functions [2, 3]. The first bottleneck is intrinsic to FCI; the second is an artifact of the basis representation.

In a series of prior papers, we developed a spectral graph theory approach to quantum chemistry in which the one-electron Hamiltonian is constructed as a sparse graph Laplacian over hydrogenic quantum numbers [4–6]. For single-electron systems (H, He^+, Li^{2+}) that graph Laplacian's extremal eigenvalue is matched to $-Z^2/2$ by construction through the scale $\kappa = -1/16$ (Paper 7); it is not a discretization converging to the S^3 Laplace–Beltrami operator (its edges never change l, so it splits into disconnected l-blocks), and the S^3 equivalence itself is Fock's continuum projection. The exact one-electron operator on the (n, l, m) labels is the diagonal hydrogenic spectrum $-Z^2/2n^2$ (equivalently, the $SO(4)$ Casimir $n^2 - 1$). The key architectural feature is that the one-electron matrix is diagonal on the (n, l, m) labels and the two-electron integrals obey the Gaunt angular-momentum selection rules, so the integral tables are sparse.

In this paper, we extend the framework to multi-electron FCI for helium ($Z = 2$, 2 electrons), lithium ($Z = 3$, 3 electrons), beryllium ($Z = 4$, 4 electrons), and the heteronuclear diatomic lithium hydride (LiH, 4 electrons). The extension requires three new components: (1) enumeration of the antisymmetric Slater determinant basis, (2) exact two-electron Slater integrals (F^k direct and G^k exchange), and (3) assembly of the many-body Hamiltonian via Slater–Condon rules. To reach the basis sizes needed for convergence, we replace the $O(N_{\mathrm{SD}}^2)$ pairwise assembly with an excitation-driven direct CI algorithm [1] that achieves $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$ scaling. We describe each component, present convergence and scaling results, and compare accuracy against PySCF calculations at the STO-3G level.

II. METHOD

A. Lattice index and Slater determinant basis

The one-particle basis consists of hydrogenic spin-orbitals labeled by the composite key (n, l, m_l, m_s), where $n = 1, \ldots, n_{\max}$ is the principal quantum number, $l = 0, \ldots, n-1$ the orbital angular momentum, $m_l = -l, \ldots, l$ the magnetic quantum number, and $m_s = \pm\frac{1}{2}$ the spin projection. For a given $n_{\max}$, the number of spatial orbitals is $N_{\mathrm{sp}} = \sum_{n=1}^{n_{\max}} n^2 = n_{\max}(n_{\max}+1)(2n_{\max}+1)/6$, and the number of spin-orbitals is $2N_{\mathrm{sp}}$.

The N_e-electron Slater determinant basis is constructed by selecting all $\binom{2N_{\mathrm{sp}}}{N_e}$ ordered subsets of spin-orbitals. Each Slater determinant $|\Phi_I\rangle$ is stored as a sorted tuple of spin-orbital indices, enabling $O(1)$ lookup

via dictionary hashing. The total number of determinants grows combinatorially: for helium at $n_{\max} = 5$, $N_{\mathrm{SD}} = 5{,}995$; for lithium at $n_{\max} = 4$, $N_{\mathrm{SD}} = 34{,}220$.

B. One-electron Hamiltonian

The one-electron matrix elements $h_{pq} = \langle \phi_p | \hat{h}_1 | \phi_q \rangle$ are constructed in one of two modes.

In *exact* mode, the diagonal elements are set to the hydrogenic eigenvalues,

$$h_{pp} = -\frac{Z^2}{2n_p^2}\,, \tag{1}$$

with all off-diagonal elements zero. This is exact for the hydrogen-like one-electron problem and provides a variationally rigorous diagonal basis.

In *hybrid* mode, the diagonal elements are given by Eq. (1), while the off-diagonal elements are taken from the graph Laplacian,

$$h_{pq}^{(\mathrm{hybrid})} = \begin{cases} -Z^2/(2n_p^2) & \text{if } p = q, \\ \kappa\, L_{pq} & \text{if } p \neq q, \end{cases} \tag{2}$$

where $L = D - A$ is the graph Laplacian of the quantum number adjacency graph and $\kappa = -1/16$ is the matched scale of prior work [4] (it fixes the one-electron extremal eigenvalue by construction; it is not a validated kinetic operator). The off-diagonal entries are a constant $+1/16$ along every lattice edge (radial $n \leftrightarrow n \pm 1$ at fixed l, m, and angular $m \leftrightarrow m \pm 1$ at fixed n, l).

[CORRECTED 2026-09-25] Only exact mode is helium's (or lithium's) Hamiltonian. The hydrogenic orbitals at $k = Z$ are eigenfunctions of $-\frac{1}{2}\nabla^2 - Z/r$, so the physical one-body matrix is exactly Eq. (1); the hybrid off-diagonal is an added operator with no derivation (and its $m \leftrightarrow m \pm 1$ part is not even rotationally invariant). Hybrid mode therefore defines a different, hop-augmented model Hamiltonian whose energies are not variational bounds on the atom. Earlier versions reported that hybrid mode "yields better accuracy for helium": it lowers the helium energy by ≈ 49 mHa, more than helium's total correlation energy (42.0 mHa), and drives lithium below its exact energy (Sec. IV B). All helium energies below are quoted with exact h_1; the hybrid values are kept only as a labeled record.

C. Two-electron integrals and Slater–Condon assembly

The two-electron repulsion integrals are computed exactly in the hydrogenic basis using Slater's decomposition into radial and angular factors [7, 8]. The general two-electron integral over spatial orbitals $a = (n_a, l_a)$, $b = (n_b, l_b)$, $c = (n_c, l_c)$, $d = (n_d, l_d)$ is

$$\langle ab|cd\rangle = \sum_k c^k(l_a m_a, l_c m_c)\, c^k(l_b m_b, l_d m_d)\, R^k(abcd)\,, \tag{3}$$

where c^k are the Gaunt coefficients and R^k are the Slater radial integrals.

The Gaunt coefficients are expressed in terms of Wigner $3j$ symbols:

$$c^k(lm, l'm') = (-1)^m \sqrt{(2l+1)(2l'+1)} \begin{pmatrix} l & k & l' \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} l & k & l' \\ -m & m-m' & m' \end{pmatrix} \tag{4}$$

The angular selection rules encoded in the $3j$ symbols—the triangle inequality $|l - l'| \le k \le l + l'$ and the parity rule $l + k + l' =$ even—produce a sparse set of nonzero coefficients that is precomputed and cached.

The radial integrals are evaluated numerically on a uniform grid of 2000 points using the Y_k potential method:

$$R^k(abcd) = \int_0^\infty \int_0^\infty \frac{r_<^k}{r_>^{k+1}}\, R_{n_a l_a}(r_1)\, R_{n_c l_c}(r_1)\, R_{n_b l_b}(r_2)\, R_{n_d l_d}(r_2)\, r_1^2 r$$

(5)

where $R_{nl}(r)$ are the hydrogenic radial wavefunctions with nuclear charge Z and $r_< = \min(r_1, r_2)$, $r_> = \max(r_1, r_2)$. The inner integral over r_2 is computed via cumulative trapezoidal sums, reducing the two-dimensional integral to a sequence of one-dimensional operations. The computed R^k values are cached to disk, making subsequent calculations effectively instantaneous.

The many-body Hamiltonian is assembled by iterating over all pairs of Slater determinants $(|\Phi_I\rangle, |\Phi_J\rangle)$ and applying the Slater–Condon rules [2]:

Diagonal ($I = J$):

$$H_{II} = \sum_{p \in I} h_{pp} + \sum_{p<q \in I} \big[\langle pq|pq\rangle - \langle pq|qp\rangle\big]\,. \tag{6}$$

Single excitation ($|\Phi_I\rangle$ and $|\Phi_J\rangle$ differ by one spin-orbital, $p \to r$):

$$H_{IJ} = (-1)^\sigma \Big[h_{pr} + \sum_{q \in I \cap J} \big[\langle pq|rq\rangle - \langle pq|qr\rangle\big] \Big]\,, \tag{7}$$

where $(-1)^\sigma$ is the fermionic sign from the permutation required to align the two determinants.

Double excitation ($|\Phi_I\rangle$ and $|\Phi_J\rangle$ differ by two spin-orbitals, $pq \to rs$):

$$H_{IJ} = (-1)^\sigma \big[\langle pq|rs\rangle - \langle pq|sr\rangle\big]\,. \tag{8}$$

Determinant pairs differing by more than two spin-orbitals produce zero matrix elements by the Slater–Condon rules and are skipped. The resulting Hamiltonian matrix is real symmetric and extremely sparse: at $n_{\max} = 5$ for helium, 2.1% of matrix entries are nonzero; at $n_{\max} = 4$ for lithium, only 0.37%.

TABLE I. Ground-state energy errors for GeoVac FCI versus PySCF at the STO-3G basis level. GeoVac results use `slater_full` two-electron integrals; all atoms use exact (physical) h_1, which for hydrogen is the exact spectrum at every $n_{\max}$ by construction. The PySCF helium result is RHF, which equals FCI for STO-3G (one spatial basis function). Exact non-relativistic energies: $E_{\mathrm{Be}} = -14.6674$ Ha [10].

System	Method	E (Ha)	Error (%)	N_{SD}
H ($Z=1$)	GeoVac (any $n_{\max}$)	−0.5000	0	—
	PySCF/STO-3G	−0.4666	6.68	1
He ($Z=2$)	GeoVac $n_{\max}=5$	−2.8476	1.93	5995
	PySCF/STO-3G	−2.8078	3.30	1
Li ($Z=3$)	GeoVac $n_{\max}=4$	−7.3987	1.06	34 220
	GeoVac $n_{\max}=5$	−7.4007	1.03	215 820
	PySCF/STO-3G	—	—	—
Be ($Z=4$)	GeoVac $n_{\max}=3$	−14.558	0.75	20 475
	GeoVac $n_{\max}=4$	−14.563	0.71	487 635

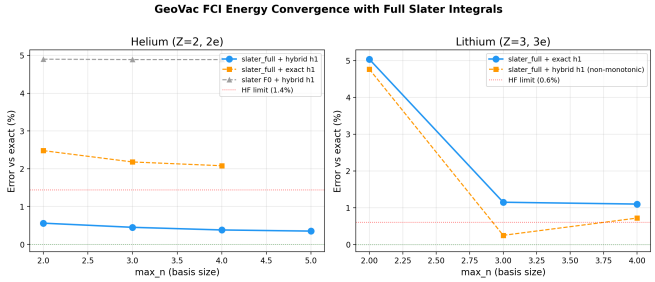


FIG. 1. Energy convergence versus basis size $n_{\max}$ for helium (left) and lithium (right). The helium curve (solid blue) is the hop-augmented hybrid model of Eq. (2), not helium; helium's physical series is in the text and Table III. Lithium uses exact h_1 (solid blue) with hybrid h_1 shown for comparison (dashed orange) and converges monotonically from above. Dotted red lines mark the Hartree–Fock limits (1.4% for He, 0.6% for Li).

D. Excitation-driven direct CI

The $O(N_{\mathrm{SD}}^2)$ pairwise determinant comparison loop of the explicit Hamiltonian assembly is replaced by an excitation-driven approach following Knowles and Handy [1]. For each determinant $|\Phi_I\rangle$, all singly and doubly excited determinants $|\Phi_J\rangle$ connected by the Hamiltonian are generated directly, and the target determinant is located via $O(1)$ dictionary lookup in the sorted-tuple index. Two-electron contributions are restricted to precomputed non-zero spatial orbital pairs, reducing the inner loop from $O(n_{\mathrm{virt}}^2)$ to $O(N_{\mathrm{nnz}}/n_{\mathrm{pairs}})$.

One-electron and two-electron lookup tables are stored as dense NumPy arrays ($n_{\mathrm{spinorb}} \leq 110$ at $n_{\max}=5$, requiring < 1 MB), avoiding the `scipy.sparse` validation overhead that dominated the hot path at $\sim$24 μs per element access. The resulting scaling is $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$ per assembly, where $N_{\mathrm{connected}}$ depends on the ERI sparsity rather than N_{SD}.

For lithium at $n_{\max}=4$, the direct CI assembly reduces the wall time from 340 s to 7.6 s (a 45× improvement) and makes $n_{\max}=5$ ($N_{\mathrm{SD}}=215{,}820$) accessible at 116 s. For beryllium at $n_{\max}=4$ ($N_{\mathrm{SD}}=487{,}635$), the direct CI solver completes in 357 s—a calculation that would require $\sim 10^{11}$ pair comparisons under the explicit assembly.

III. RESULTS

All calculations use atomic units (hartree for energy, bohr for length). Exact non-relativistic energies are $E_{\mathrm{He}} = -2.9037$ Ha [9] and $E_{\mathrm{Li}} = -7.4781$ Ha [10]. Hartree–Fock limits are $E_{\mathrm{He}}^{\mathrm{HF}} = -2.8617$ Ha and $E_{\mathrm{Li}}^{\mathrm{HF}} = -7.4327$ Ha [11].

Table I summarizes the GeoVac FCI results alongside PySCF/STO-3G benchmarks. GeoVac achieves 1.93% error for helium, 1.03% for lithium (at $n_{\max}=5$), and 0.71% for beryllium (at $n_{\max}=4$), compared to 3.30% and 6.68% for PySCF/STO-3G on helium and hydrogen, respectively; the hydrogen entry is exact by construction, since the physical h_1 is the hydrogenic spectrum.

A note on the scope of this comparison is warranted. STO-3G is a minimal Gaussian basis introduced by Hehre, Stewart, and Pople in 1969 [16]; it is not used in production quantum chemistry, where correlation-consistent bases such as cc-pVDZ or cc-pVTZ are standard. The comparison is not at matched size: STO-3G has one function for helium, against 55 spatial orbitals at $n_{\max}=5$. Against the strongest baselines it is unfavourable: the GeoVac helium energy stays above the Hartree–Fock limit at every $n_{\max}$ reached, whereas a cc-pVDZ FCI (five functions) already recovers correlation below it, and at equal qubit count the GeoVac helium energy lies tens of mHa above cc-pVDZ and cc-pVTZ (Paper 14).

Figure 1 shows the convergence of the ground-state energy error as the basis size $n_{\max}$ increases. For helium with the physical (exact) h_1, the error decreases monotonically: 2.43% ($n_{\max}=2$) → 2.10% ($n_{\max}=3$) → 1.99% ($n_{\max}=4$) → 1.93% ($n_{\max}=5$), and flattens *above* the Hartree–Fock limit of 1.4%. A fixed-exponent ($k=Z$) set of hydrogenic bound states omits the continuum and is not complete, so these calculations do not reach the Hartree–Fock limit, let alone recover correlation beyond it. (The earlier text quoted the hybrid series 0.50 → 0.26% and read it as correlation beyond mean-field; the hybrid energies belong to the hop-augmented model of Sec. IV B, not to helium.) For lithium with exact h_1, the error decreases monotonically: 5.00% ($n_{\max}=2$) → 1.12% ($n_{\max}=3$) → 1.06% ($n_{\max}=4$). The lithium error at $n_{\max} \geq 3$ (1.06–1.12%) remains *above* the Hartree–Fock limit of 0.61%, so—like helium—the lithium FCI does not recover correlation energy beyond mean-field theory at the basis sizes reached here.

Figure 2 shows the scaling of the Hamiltonian con-

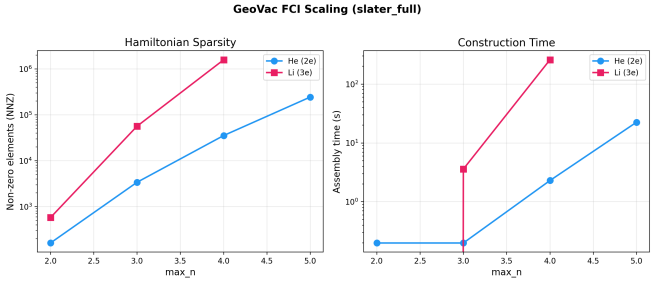


FIG. 2. Scaling of the FCI Hamiltonian with basis size. Left: number of nonzero matrix elements (NNZ) versus $n_{\max}$. Right: wall-clock assembly time versus $n_{\max}$. Both axes are logarithmic. Fitted to the exact-h_1 NNZ of Table III, the growth exponent is $\alpha \approx 1.8$ for He and $\alpha \approx 1.6$ for Li (sub-quadratic in N_{SD}).

struction. The number of nonzero elements grows sub-quadratically in the number of Slater determinants, with fitted exponents NNZ $\sim N_{\mathrm{SD}}^{1.8}$ for helium and $N_{\mathrm{SD}}^{1.6}$ for lithium (Table III). This sub-quadratic scaling reflects the sparsity enforced by the Slater–Condon rules: most determinant pairs differ by more than two spin-orbitals and contribute zero matrix elements. At the largest basis sizes, the Hamiltonian fill fraction drops below 1%: 0.17% for Li at $n_{\max} = 5$ (78,025,400 nonzeros out of $215{,}820^2$ entries) and 0.05% for Be at $n_{\max} = 4$; for He at $n_{\max} = 5$ it is 2.1% (759,739 nonzeros out of $5{,}995^2$ entries).

With the direct CI assembly (Sec. II D), wall times range from sub-second for small bases to 7.6 s for Li at $n_{\max} = 4$ and 357 s for Be at $n_{\max} = 4$. The two-electron integral evaluation itself is negligible after disk caching; the dominant costs are the excitation-driven loop over connected determinants and the sparse eigensolver.

A. Heteronuclear diatomic: LiH

As the first heteronuclear test case, we apply the molecular FCI framework to lithium hydride (LiH, $Z_{\mathrm{Li}} = 3$, $Z_{\mathrm{H}} = 1$, 4 electrons). The molecular Hamiltonian is assembled from two atomic hydrogenic lattices ($n_{\max} = 3$ per atom) using the `MolecularLatticeIndex` class, with cross-nuclear attraction evaluated via the electrostatic potential of the Mulliken-approximated orbital density and same-atom two-electron integrals supplemented by 36 cross-atom two-electron integrals—18 direct Coulomb entries from the Fourier convolution formula

$$J_{AB}(R) = \frac{2}{\pi} \int_0^\infty \tilde{\rho}_A(q)\, \tilde{\rho}_B(q)\, \frac{\sin(qR)}{qR}\, dq\,, \qquad (9)$$

where $\tilde{\rho}(q)$ is the momentum-space orbital density of Ref. [6], Eq. (32)—and 18 cross-atom exchange entries from the Mulliken approximation

$$K_{AB}(R) = S_{AB}(R)^2\, \frac{F^0(a,a;Z_A) + F^0(b,b;Z_B)}{2}\,, \qquad (10)$$

where S_{AB} is the two-center overlap integral and F^0 denotes same-atom direct Slater integrals. The combined spin-orbital basis has 56 spin-orbitals, yielding $\binom{56}{4} = 367{,}290$ Slater determinants; each geometry assembles in approximately 75 s. The results below are reported at $n_{\max} = 2$ (20 spin-orbitals, 4,845 determinants), the size at which the physical one-electron Hamiltonian has been run.

a. Basis set superposition error. When two atom-centered bases are combined into a shared molecular spin-orbital space, the FCI solver gains variational access to basis functions from both centers, artificially lowering the molecular energy relative to the isolated-atom reference. The Boys–Bernardi counterpoise (CP) correction [13]

$$\mathrm{BSSE} = \left[E_{\mathrm{Li}}^{\mathrm{ghost}} + E_{\mathrm{H}}^{\mathrm{ghost}}\right] - \left[E_{\mathrm{Li}}^{\mathrm{own}} + E_{\mathrm{H}}^{\mathrm{own}}\right], \qquad (11)$$

where ghost-orbital energies are computed with the full molecular basis but zero nuclear charge on the other center, gives -0.088 Ha at $n_{\max} = 2$ with the physical one-electron Hamiltonian, the same at every R from 1.5 to 8 bohr, comparable to the experimental binding energy (0.0924 Ha).

b. Results. Table II gives the LiH potential energy surface at $n_{\max} = 2$ with the physical one-electron Hamiltonian (hydrogenic diagonal plus cross-nuclear attraction, with the STO-overlap bridges as the only off-diagonal one-electron terms). The energy falls monotonically as R decreases, with no equilibrium geometry, and at $R = 8$ bohr it still lies 0.22 Ha below the separated-atom energy, so the surface does not dissociate to the atoms and no binding energy is extracted. Consistent with the graph-concatenation analysis of the companion molecular paper, the structural cause is the R-independence of each atom's one-electron block (its hydrogenic diagonal), which supplies no orthogonalization kinetic repulsion as the centers approach.

c. Error budget. A systematic diagnostic decomposition—zeroing cross-nuclear attraction, cross-atom V_{ee}, and bridge terms independently—confirmed that cross-nuclear attraction is computed exactly via the Fourier electrostatic potential (Newton's shell theorem), matching the analytical two-center integral to 2×10^{-15} Ha. Extending cross-atom direct Coulomb integrals to all orbital pairs (monopole approximation) shifts the energy by only 0.5 mHa, so missing integrals do not explain the surface; the absence of a minimum follows from the R-independent one-electron blocks, which a larger $n_{\max}$ does not change.

IV. DISCUSSION

A. Architectural advantages

The central architectural feature of this approach is the *relational lattice index*: each Slater determinant is a tuple of spin-orbital keys (n, l, m_l, m_s), and the Hamiltonian

TABLE II. LiH potential energy surface at $n_{\max} = 2$ per atom with the physical one-electron Hamiltonian (balanced configuration). Experimental values: $R_{\rm eq} = 3.015$ bohr, $D_e = 0.0924$ Ha [12].

R (bohr)	$E_{\rm mol}$ (Ha)
1.5	−8.3793
2.5	−8.0592
3.015	−7.9899
4.0	−7.9312
5.0	−7.8877
8.0	−7.8258
separated atoms	−7.601

TABLE III. Detailed convergence data for helium, lithium, and beryllium, all with exact (physical) h_1 (helium rows corrected 2026-09-25 from the hybrid values −2.889303, −2.892932, −2.895227, −2.896298; NNZ is that of the exact-h_1 matrix). $N_{\rm SD}$ is the number of Slater determinants, NNZ the number of nonzero Hamiltonian entries, and t the wall-clock assembly time using direct CI (Sec. II D).

Z	$n_{\max}$	$N_{\rm SD}$	NNZ	E (Ha)	Err. (%)	t (s)
2	2	45	139	−2.833222	2.43	<1
2	3	378	5 234	−2.842724	2.10	<1
2	4	1 770	82 910	−2.846005	1.99	<1
2	5	5 995	759 739	−2.847551	1.93	1.6
3	2	120	604	−7.104490	5.00	<1
3	3	3 276	111 192	−7.394597	1.12	<1
3	4	34 220	4 385 956	−7.398718	1.06	7.6
3	5	215 820	78 025 400	−7.400705	1.03	116
4	3	20 475	1 224 507	−14.558052	0.75	2.9
4	4	487 635	118 132 111	−14.562659	0.71	357

construction reduces to relational operations (set differences, index lookups) over these keys. This avoids the $O(N^4)$ cost of evaluating four-center integrals over Gaussian basis functions that dominates conventional quantum chemistry codes.

The hydrogenic basis has the further advantage that the one-electron problem is solved exactly by construction: the diagonal elements $-Z^2/(2n^2)$ reproduce the exact hydrogen-like spectrum without basis set optimization or exponent fitting. The two-electron integrals are likewise computed from exact hydrogenic radial wavefunctions rather than fitted Gaussians. The basis set error is that of the fixed-exponent bound-state basis itself: truncation at finite $n_{\max}$ and, more fundamentally, the omission of the continuum (Sec. IV B).

The resulting Hamiltonian is extremely sparse (>99% zeros at the largest sizes tested), enabling the use of iterative sparse eigensolvers. The eigensolver cost is negligible compared to the assembly cost in all cases tested (Table III).

B. Known limitations

Hybrid h_1 overshoot for lithium. When the hybrid one-electron Hamiltonian [Eq. (2)] is used for lithium, the energy at $n_{\max} = 3$ drops to −7.4966 Ha, which is 0.25% *below* the exact non-relativistic value −7.4781 Ha. At $n_{\max} = 4$, the overshoot worsens to −7.5318 Ha (0.72% below exact). This violation of the variational principle is the visible symptom of a general fact: the hybrid off-diagonal terms are not part of the atom's Hamiltonian (Sec. II B), so nothing bounds the hybrid energy from below by the exact one. The exact h_1 mode [Eq. (1)] is the physical Hamiltonian: the lithium energy remains above the exact value at all basis sizes, converging monotonically from 5.00% to 1.03%.

Hybrid h_1 for helium [CORRECTED 2026-09-25]. Earlier versions concluded that helium, which does not overshoot at $Z = 2$, should keep the hybrid mode because it "produces lower errors" (0.29% versus 1.99% at $n_{\max} = 4$). That is the same non-physical term, merely not yet below the exact line: the hop lowers the helium energy by 49–56 mHa at $n_{\max} = 2$–5, more than helium's total correlation energy (42.0 mHa), and the hybrid ground state evaluated on helium's physical Hamiltonian lies *higher* (2.23% at $n_{\max} = 4$) than the exact-h_1 ground state (1.98%). Applied to the helium-like sequence, the hybrid energy falls below the exact energy for nuclear charge $Z \lesssim 1.8$–1.9 (H^- at $Z = 1$ is over-bound by 21%); the exact-h_1 calculation is variational at every Z. The hybrid results are therefore withdrawn as helium energies.

Continuum incompleteness. With exact h_1 the helium error flattens at ≈1.9%, above the Hartree–Fock limit. The basis is the discrete hydrogenic spectrum at the fixed exponent $k = Z$, which omits the continuum and is not complete; a variational or Sturmian exponent is the standard remedy.

Construction time scaling. The excitation-driven direct CI assembly (Sec. II D) reduces the scaling from the $O(N_{\rm SD}^2)$ pairwise comparison of the original implementation to $O(N_{\rm SD} \times N_{\rm connected})$. For lithium at $n_{\max} = 4$, this reduces the wall time from 340 s to 7.6 s (a 45× improvement), and makes $n_{\max} = 5$ ($N_{\rm SD} = 215{,}820$) accessible at 116 s. For beryllium at $n_{\max} = 4$ ($N_{\rm SD} = 487{,}635$), the assembly completes in 357 s. Extension to $n_{\max} = 6$ or five-electron systems will likely require further algorithmic improvement (e.g., string-based factorization or selected CI).

Remaining basis set error. The helium error of 1.93% at $n_{\max} = 5$, lithium error of 1.03% at $n_{\max} = 5$, and beryllium error of 0.71% at $n_{\max} = 4$ remain above chemical accuracy (∼1 kcal/mol ≈ 0.05% for these systems). Larger $n_{\max}$ alone will not reach it: the helium series flattens near 1.9% because the fixed-k hydrogenic bound states omit the continuum. A variational or Sturmian exponent is required.

LiH: no equilibrium geometry. For the heteronuclear diatomic LiH, the surface is monotonically attractive and

does not return to the separated-atom energy at large R. A systematic diagnostic decomposition confirmed that cross-nuclear attraction is computed exactly via Newton's shell theorem (the Fourier electrostatic potential matches the analytical two-center integral to machine precision), and that adding all-orbital cross-atom direct Coulomb integrals (monopole approximation) shifts the energy by only 0.5 mHa—ruling out missing integrals as the error source. The structural cause is the R-independence of each atom's one-electron block, which supplies no orthogonalization kinetic repulsion; increasing $n_{\max}$ does not remove it, and the natural two-center geometry (prolate spheroidal, Paper 11 [15]) is the route that does.

V. HETERONUCLEAR DIAGNOSTIC ARC (V0.9.11–V0.9.30)

Following the initial LiH implementation (v0.9.9), we undertook a systematic exploration of alternative molecular Hamiltonians for the heteronuclear case. Table IV summarizes the complete diagnostic arc. Every version targets LiH at $n_{\max} = 3$ (367,290 Slater determinants) unless noted otherwise.

The diagnostic arc establishes three conclusions:

1. **The atom-centered Mulliken/Fourier baseline (v0.9.13) is the best atom-centered result of the arc** (on its hop-augmented operator). All subsequent variants either fail completely or degrade accuracy.

2. **Single-p_0 Sturmian bases cannot bind heteronuclear molecules** (v0.9.20–v0.9.28): at the shared p_0 determined by the heavier atom, the lighter center's $1s$ orbital is pushed into positive energy by $p_0^2/2 - Z_B p_0 > 0$. None of the atom-centered cross-nuclear corrections tried in those versions compensated; for the genuine cross-n Sturmian overlap, outside this construction, binding is not excluded (Paper 8, Remark).

3. **Molecular Sturmians via prolate spheroidal separation (v0.9.29) compute the correct β_k eigenvalues** (H_2^+: 0.6% error), but projecting these molecular β_k back onto atom-centered bases (v0.9.30) introduces catastrophic scale mismatches. Using the molecular Sturmian functions directly as the FCI basis (v0.9.31–v0.9.33, Sec. V B) gave 12–25% errors in the companion molecular paper; in its shared-p_0, single-n form it falls under Paper 8's structural theorem, whose eigenvalues are R-independent [14].

A. Molecular Sturmian basis: prolate spheroidal implementation

The `compute_molecular_sturmian_betas()` function (v0.9.29) computes molecular Sturmian β_k eigenvalues by separating the two-center Coulomb Sturmian equation in prolate spheroidal coordinates (ξ, η, φ) with foci at the two nuclei.

a. Angular eigenvalue. The angular separation constant $A_{m,n_{\rm sph}}(c, b)$ is computed via matrix diagonalization of the Legendre expansion of the spheroidal equation, where $c = p_0 R/2$ is the spheroidal parameter and $b = (Z_A - Z_B)R/2$ encodes charge asymmetry. Results match `scipy.special.obl_cv` to $\sim 10^{-14}$ for the homonuclear case ($b = 0$).

b. Radial eigenvalue. The top radial eigenvalue $L_{\rm rad}(c, a)$ is computed via self-adjoint finite-difference discretization with Neumann boundary conditions at $\xi = 1$ for $m = 0$ (critical for the ground state). The parameter $a = \beta_k(Z_A + Z_B)R/2$ encodes the molecular Sturmian eigenvalue.

c. Root-finding. For each orbital $(m, n_{\rm sph}, n_{\rm rad})$, β_k is found via `scipy.optimize.brentq` on the matching condition $A_{\rm ang}(c, b) + L_{\rm rad}(c, a(\beta_k)) = 0$.

d. H_2^+ validation. At $R = 2.0$ bohr, the prolate spheroidal solver gives the $1\sigma_g$ energy $E = -1.096$ Ha (exact: -1.103 Ha, 0.6% error), validating the numerical implementation.

e. LiH results. At $R = 3.015$ bohr with $p_0 = \sqrt{10}$: 10 molecular orbitals found. 1σ: $\beta = 1.033$ (Li $1s$ core); 2σ: $\beta = 1.950$ (Li-H bonding); 1π: $\beta = 1.976$. All $n = 3$ orbitals have $\beta > 2.3$.

B. Molecular orbital FCI architecture (v0.9.31 specification)

The v0.9.31 solver uses the molecular Sturmian functions directly as the FCI basis, bypassing all atom-centered approximations; it gave 12–25% errors, and its shared-p_0 D-matrix off-diagonal places it within Paper 8's single-n theorem. The theoretical foundation is given in the companion theory paper [14], Sec. X.

a. Data structures.

- `MolecularSturmianBasis`: stores $(m, n_{\rm sph}, n_{\rm rad}, \beta_k)$ for each MO. At $n_{\max} = 3$: 10 spatial MOs, 20 spin-orbitals.

- `h1_mo[i,j]`: dense 10×10 one-electron matrix. Diagonal: $\varepsilon_k = \frac{1}{2}p_0^2 - p_0^2/\beta_k$ (equals $-p_0^2/2$ only in the degenerate case $\beta_k = 1$). Off-diagonal: D-matrix elements weighted by nuclear charges (same-m block-diagonal).

- `eri_mo[i,j,k,l]`: dense $10 \times 10 \times 10 \times 10$ two-electron integral tensor. Computed from projected atom-centered Slater integrals via prolate spheroidal overlap coefficients.

TABLE IV. Heteronuclear diagnostic arc (v0.9.11–v0.9.30). All calculations: LiH, $n_{\max} = 3$, 367,290 SDs, with the hop-augmented one-body operator of these versions; with the physical h_1 the surface has no equilibrium (Sec. III A), so the D_e^{CP} and R_{eq} columns are records of that operator, not LiH results. D_e^{CP} is Boys–Bernardi counterpoise-corrected binding energy. Experimental: $D_e = 0.092$ Ha, $R_{\mathrm{eq}} = 3.015$ bohr.

Version	Architecture	D_e^{CP} (Ha)	R_{eq} (bohr)	Status	Root Cause
v0.9.11	Mulliken baseline	0.093	$\sim$2.5	PARTIAL	R_{eq} too short
v0.9.12	Löwdin 4-index ERI	—	—	NEGATIVE	Catastrophically wrong
v0.9.13	Mulliken exchange K	—	$\sim$2.5	PARTIAL	R_{eq} shifted
v0.9.16	D-matrix geometric-mean	-1.237	none	NEGATIVE	Coupling 5$\times$ too weak
v0.9.17	SW off-diagonal only	-1.060	none	NEGATIVE	Missing diagonal cross-nuclear
v0.9.18	Hybrid (Fourier diag + SW off-diag)	0.143	$<$2.0	PARTIAL	SW overbinding at short R
v0.9.20	Sturmian + Fourier hybrid	diverges	—	NEGATIVE	Fourier–Sturmian inconsistency
v0.9.21	p_0-consistent Sturmian	unbound	—	NEGATIVE	Single p_0 unbinds H (+1.85 Ha)
v0.9.22	Atom-dependent p_0	2.556	none	NEGATIVE	28$\times$ overbinding (cap saturates)
v0.9.23	Shell-radius R_{eff}	unbound	—	NEGATIVE	Double-counts orbital extent
v0.9.24	Term decomposition (diagnostic)	—	$\sim$2.5	—	$V_{\mathrm{cross},B}$ drives R_{eq}
v0.9.25	Cross-atom V_{ee} (Ohno-Klopman)	unbound	—	NEGATIVE	Over-repulsive for $l > 0$ pairs
v0.9.26	Angular-weighted Ohno-Klopman	unbound	—	NEGATIVE	Negligible effect (-85 μHa)
v0.9.27	SO(4) CG cross-center V_{ee}	unbound	—	NEGATIVE	$J_{\mathrm{cross}} > J_{\mathrm{same}}$ (artifact)
v0.9.28	Molecular β_k (node weights)	unbound	—	NEGATIVE	$\beta_k = 1.312 < 1.582$ threshold
v0.9.29	Prolate spheroidal β_k	—	—	PARTIAL	H_2^+ 0.6% error; MO$\rightarrow$atom map ad hoc
v0.9.30	MO-projected β_k	-42.5	—	NEGATIVE	Beta scale mismatch (25–170$\times$)

- Slater determinant basis: $\binom{20}{4} = 4{,}845$ determinants for 4 electrons in 20 spin-orbitals (LiH).

b. Function signatures.

- `compute_molecular_sturmian_betas(Z_A, Z_B, R, p0, nmax) -> List[MolecularOrbital]`: prolate spheroidal β_k solver.
- `build_mo_h1(mos, p0, gamma, Z_A, Z_B) -> ndarray`: one-electron Hamiltonian in MO basis.
- `build_mo_eri(mos, Z_A, Z_B) -> ndarray`: two-electron integrals via projected Slater integrals.
- `solve_mo_fci(Z_A, Z_B, R, nmax) -> (energy, civec)`: top-level driver with p_0 self-consistency.

c. Expected computational cost. With $n_{\max} = 3$: 10 MOs per spin, 20 spin-orbitals, $\binom{20}{4} = 4{,}845$ Slater determinants for 4 electrons. Based on direct CI scaling at comparable N_{SD} (cf. He at $n_{\max} = 5$: 5,995 SDs in 1.6 s), estimated runtime is < 2 s per geometry per self-consistency iteration. The β_k computation via prolate spheroidal root-finding adds < 1 s per iteration (10 brentq solves). With ~ 10 self-consistency iterations, total runtime per geometry point is estimated at < 30 s.

VI. CONCLUSION

We have demonstrated that full configuration interaction in a hydrogenic basis on the S^3 label set, with the diagonal hydrogenic one-body term and exact Slater two-electron integrals, achieves 1.93% accuracy for helium, 1.03% for lithium, and 0.71% for beryllium with each atom's physical Hamiltonian—all with monotonic basis set convergence and all limited by the incompleteness of a fixed-exponent hydrogenic bound-state basis (the helium and lithium errors sit above their Hartree–Fock limits, so the comparison with correlation-consistent Gaussian bases is unfavourable; only the one-function STO-3G basis is bettered). The excitation-driven direct CI algorithm reduces assembly cost from $O(N_{\mathrm{SD}}^2)$ to $O(N_{\mathrm{SD}} \times N_{\mathrm{connected}})$, enabling calculations with nearly 500,000 determinants. For atoms every two-electron integral is one-center and reduces, by Slater's radial–angular decomposition, to R^k radial integrals and Gaunt coefficients; with the Slater–Condon rules this yields Hamiltonians with sub-quadratic nonzero growth and >99% sparsity at the largest sizes tested. We demonstrate the extension to heteronuclear molecules with LiH, the first diatomic calculation in the framework: the R-independent one-electron block of each atom yields a monotonically attractive surface with no equilibrium geometry that does not return to the separated-atom energy at large R, and a systematic diagnostic decomposition—zeroing cross-nuclear, cross-atom V_{ee}, and bridge terms independently—rules out missing integrals as the cause. For the equilibrium geometry problem—determining R_{eq} from first principles—the prolate spheroidal lattice of Paper 11 [15] provides the principled resolution, reaching machine precision for H_2^+ with zero fitted parameters by working in the label set and channel structure of the natural two-center coordinate system, where the Schrödinger equation separates.

ACKNOWLEDGMENTS

Computational resources were provided by personal hardware.

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a. Note added v2.6.0. The numerical Slater integrals (Sec. III, 2000-point grid) have been replaced by exact rational values (s-pair F^0 via the Parseval form of Paper 7, Sec. V; general F^k by closed form), so the entire Hamiltonian matrix is rational. With the physical (exact) h_1, the singlet $M_L = 0$ CI for He lies 2.42, 2.09, 1.98, 1.93, 1.90, 1.88% above exact at $n_{\max} = 2, \ldots, 7$ (1,218 configurations at $n_{\max} = 7$), agreeing with the determinant route of Table III to ≤ 0.2 mHa.

[CORRECTED 2026-09-25] This note previously reported a "graph-native" FCI reaching 0.25% ($n_{\max} = 5$), 0.23% (6) and 0.216% (7), obtained with the hybrid h_1 of Eq. (2). Those are energies of the hop-augmented model Hamiltonian (Sec. IV B), not of helium, and are withdrawn as helium results. The note went on to call a variational-k diagonal variant (1.41% at $n_{\max} = 7$) "7× worse", "demonstrating that the graph Laplacian's off-diagonal couplings are the dominant source of one-body correlation", and to attribute a ∼0.14% floor to the cusp. In the $k = Z$ basis the physical one-body operator has no off-diagonal elements, so there is no one-body correlation for the hop to supply; what the hop does is compensate most (≈ 86% at $n_{\max} = 4$–5) of the continuum-incompleteness error of the fixed-k basis. The cusp attribution is withdrawn with it. FCI basis invariance under orbital rotation (Sprint 3D) holds for either one-body operator and does not discriminate between them.